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Palaeomagnetism of iron-formation from the late Palaeoproterozoic Frere Formation, Earaheedy Basin, Western Australia: palaeogeographic and tectonic implications

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Abstract

Oriented cores of unweathered Frere Formation (1.9–1.8 Ga), including shaly and peloidal iron-formation, were obtained from two holes drilled to 116 m depth on opposite limbs of an anticline in the southern Earaheedy Basin at the northeastern margin of the Yilgarn Craton. Anisotropy of magnetic susceptibility measurements indicated that the Frere Formation in this area was unaffected by regional metamorphism. The predominant magnetic mineral in the iron-formation is hematite but a small amount of magnetite also is present in some samples. Stepwise thermal demagnetisation was conducted for 263 specimens. Many specimens acquired spurious components of magnetisation after treatment to 300–500 °C, ascribable to the oxidation of siderite to magnetite on heating. However, other specimens displayed a stable, high-temperature (680 °C), dual-polarity component A carried by hematite that gave a bedding dip-corrected direction of $D = 314.3^\circ$, $I = -24.8^\circ$ ($K = 38.5$, $\alpha_{95} = 2.6^\circ$, $n = 78$). A fold test was positive at >99% level of confidence, indicating acquisition of remanence prior to Stanley folding in the Earaheedy Basin that occurred at ~1.78 Ga near the end of Capricorn orogenesis. The presence of polarity reversals in the stratigraphic section from one drill hole and within some specimens argues for the identification of a dipole field axis and the sufficient averaging of secular variation to define a palaeopole for the Frere Formation, located at 45.2°S, 220.0°E (dp = 1.3°, dm = 2.4°). Component A is ascribed to a chemical remanent magnetisation (CRM) acquired soon after deposition, and indicates a palaeolatitude of $13.0 \pm 1.5^\circ$ for the Frere Formation. The Frere pole plots adjacent to, but is still significantly different from, late to post-Capricorn Orogeny overprint poles from the Hamersley Province of the southern Pilbara. However, vertical-axis relative anticlockwise rotation of 10–15° for the Frere pole brings it close to these Pilbara overprint poles. The data imply that the Earaheedy Basin and possibly the Yilgarn Craton underwent clockwise rotation of 10–15° relative to the Pilbara Craton during Stanley folding. This finding accords with evidence for tectonic settings marked by sinistral strike-slip deformation adjacent to the Yilgarn–Pilbara cratonic boundary during deposition and subsequent folding in the Earaheedy Basin. The data also imply that the West Australian and North Australian cratons were in close proximity by ~1.8 Ga.
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Keywords: Western Australia; Palaeoproterozoic; Capricorn Orogen; Earaheedy Basin; Iron-formations; Palaeomagnetism

1. Introduction

This study of the late Palaeoproterozoic Frere Formation in the Earaheedy Basin at the northeastern margin of the Yilgarn Craton is part of a wide investigation of the palaeomagnetism and rock magnetism

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of Proterozoic iron-formations. The Frere Formation includes peloidal iron-formation (Hall and Goode, 1978; Goode et al., 1983; Bunting, 1986) that is comparable to Superior-type iron-formation from the late Palaeoproterozoic Gunflint Formation in the Superior Province, Ontario (Floran and Papike, 1975), and the broadly correlative Sokoman Formation in the Labrador Trough, Québec (Dimroth and Chauvel, 1973).

Despite the economic importance of Palaeoproterozoic iron-formations, surprisingly little information is available on their original components of magnetisation and palaeolatitudes of deposition. This lack of information has come about because most studies of iron-formations have investigated the origin of secondary iron ores and so have identified overprint components (e.g. Symons, 1967a,b; Porath, 1967; Porath and Chamalaun, 1968; Purucker, 1983; Li et al., 1993). Other studies have focused on magnetite-rich iron-formations, which are strongly anisotropic and exhibit considerable deflection of remanence away from the inducing field toward the bedding plane (e.g. Schmidt and Clark, 1994). Few palaeomagnetic studies of iron-formations have sought original components. Two palaeomagnetic results interpreted as depositional palaeolatitudes, for the Gunflint Formation (69° palaeolatitude; Symons, 1966) and the Sokoman Formation (3° palaeolatitude; Seguin, 1976), are from studies carried out before the widespread use of modern palaeomagnetic analytical techniques or which did not conduct complete or stepwise demagnetisation and field tests.

The present study is the first palaeomagnetic investigation carried out in the Capricorn Orogen. It provides information on the palaeolatitude of the Frere Formation, the palaeo-orientation of the Earaheedy Basin during iron-formation deposition, and the tectonic settings at the Yilgarn–Pilbara cratonic boundary during deposition and folding in the Earaheedy Basin.

2. Geological setting

The Earaheedy Basin covers ~30,000 km^2 in the southeastern part of the Capricorn Orogen (Fig. 1; Jones et al., 2000; Pirajno et al., 2000). The Earaheedy Group, the sole lithostratigraphic group in the basin, comprises a package of shallow-water marine siliciclastic and chemical sedimentary rocks up to 5 km thick that is divisible into the Tooloo and Minningarra subgroups (Table 1). The Tooloo Subgroup rests unconformably on Archaean granites and greenstone-belt rocks of the Yilgarn Craton and Palaeoproterozoic sedimentary and volcanic rocks of the Yerrida Basin in the northwest. Jones et al. (2000) and Pirajno and Adamides (2000) interpreted the preserved Earaheedy Basin as the coastal to nearshore portion of a passive continental margin along the northeastern Yilgarn Craton, with deeper water toward the present north. The Earaheedy Basin and the older Yerrida, Bryah and Padbury basins in the southeastern Capricorn Orogen (Pirajno and Occhipinti, 2000) form the Palaeoproterozoic (~2.2–1.8 Ga) 'Nabberu province' of Gee (1990).

The Frere Formation of the Tooloo Subgroup extends throughout the Earaheedy Basin and marks an episode of widespread ferric oxide precipitation (Hall and Goode, 1978; Goode et al., 1983; Bunting, 1986). The iron-formation commonly has a granular peloidal

Table 1
Stratigraphy of the late Palaeoproterozoic (1.9–1.8 Ga) Earaheedy Group, Earaheedy Basin, Western Australia (after Bunting, 1986; Hocking et al., 2000; Jones et al., 2000)

Earaheedy Group
Minningarra Subgroup
Mulgarra Sandstone (~100+ m): Medium-grained quartz arenite; minor siltstone. Unconformably overlain by Bangemall Group
Kulele Limestone (~300 m): Oolitic, pisolitic, stromatolitic and brecciated limestone; calcarenite; sandstone; siltstone; rare dolostone
Wongawol Formation (~1500 m): Arkosic and lithic sandstone; siltstone; minor carbonate in upper part
Chiall Formation (~1000 m): Sandstone; kaolinitic siltstone; silty sandstone. Comprises the Princess Ranges, Wandiwarra and Karri Karri members
Tooloo Subgroup
Windidda Formation (~600 m): Laminated and stromatolitic dolostone and limestone; interbedded mudstone and shale. Constitutes a lateral facies equivalent of the Frere Formation
Frere Formation (~600 m): Peloidal, shaly and banded iron-formation; chert; shale; minor carbonate. Conformable on the Yelma Formation
Yelma Formation (≤1000 m): Quartz arenite; shale; stromatolitic carbonate; chert; conglomerate. Includes the Sweetwaters Well and Yadgimurrin members. Unconformable on Archaean basement and Palaeoproterozoic Yerrida Group

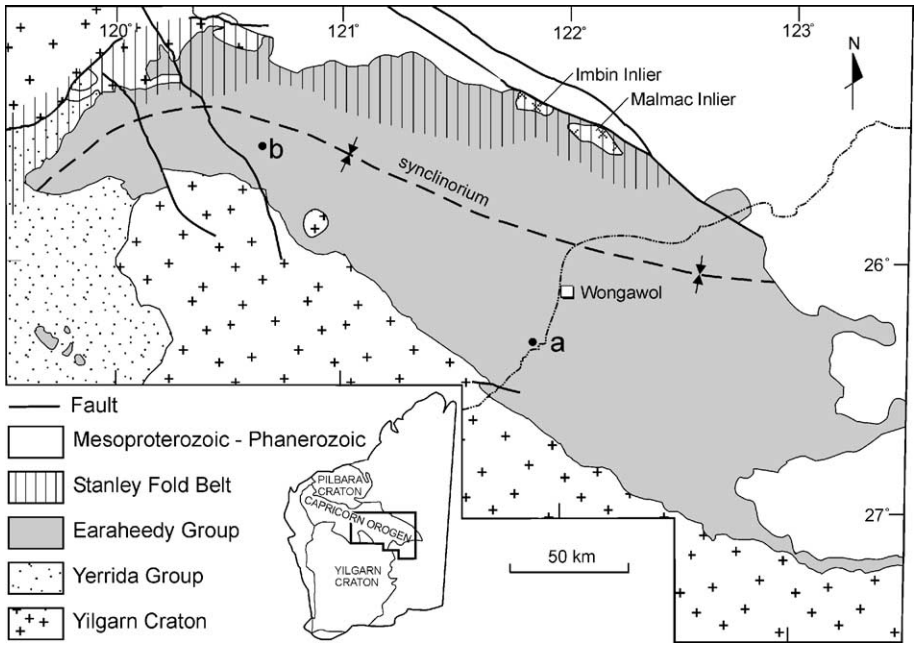


Fig. 1. Simplified geological map of the Earaheedy Basin, showing the distribution of the Palaeoproterozoic Earaheedy Group unconformably on the Archaean Yilgarn Craton, the Palaeoproterozoic Yerrida Group, the Malmac Inlier and the ~2-Ga Imbin Inlier (after Jones et al., 2000). a, Locality of drill holes DDH #1 and DDH #2 of this study in the southern Earaheedy Basin near the Gunbarrel Highway. b, Locality of Renison Gold Corporation drill holes TDH-26 and TDH-28. The map of Western Australia shows the main tectonic elements of the region.

texture and locally is oolitic or banded, and occurs in three major intervals separated by shale and minor carbonate (Jones et al., 2000; Pirajno et al., 2000). In the northwestern part of the basin the iron-formation contains >20% Fe, with local secondary enrichment to 66% Fe. The Frere Formation represents one of the few occurrences of Palaeoproterozoic Superior-type iron-formation to be recognised outside North America.

The age of the Earaheedy Group is poorly constrained. A maximum limit is provided by an age of 1990 ± 6 Ma for rhyodacite from the Imbin Inlier (Fig. 1; Nelson, 2001). Pb–Pb data for stromatolitic carbonates from the Yelma Formation at the base of the Earaheedy Group gave errorchron ages of 2008 ± 68 Ma and 1946 ± 71 Ma (Russell et al., 1994). Jones et al. (2000) noted that isotopic ages for sedimentary rocks from the Earaheedy Group and mineralisation in the basin cluster around 1.8–1.7 Ga. Grey (1994) concluded from stromatolite taxa that the age of the Earaheedy Group may be between 1.9 and 1.8 Ga. Comparative biogeology supports an age of 1.9–1.8 Ga for the Earaheedy Group: microfossils from the Frere Formation belong to taxa previously recorded in the Gunflint Formation in Canada (Walter et al., 1976; Awramik and Walter, 1978; Tobin, 1990), from which a euhedral volcanogenic zircon gave a U–Pb age of 1878 ± 2 Ma (Fralick et al., 1998).

Deformation in the Earaheedy Basin occurred during Stanley folding, which produced a synclinorium trending broadly east–west (Fig. 1). In the southern part of the basin (‘Kingston Platform’ of Gee, 1990), the rocks display some gentle folds but no obvious sign of metamorphism. These folds commonly have steeper south-dipping limbs and some are associated with axial-plane reverse faults, suggesting monoclinal draping over basement faults (Bunting, 1986). Within the Stanley Fold Belt in the northern part of the basin, the rocks show a northward increase in the intensity of folding and the development of metamorphic muscovite in pelitic rocks, as marked by progression from slaty cleavage to phyllite to muscovite schist

(Bunting, 1986). In the Frere Formation, finely divided hematite, interpreted as primary, is progressively replaced by platy and specular hematite and magnetite with increasing deformation and grade of metamorphism (Goode et al., 1983).

According to Jones et al. (2000), the Earaheedy Basin appears to have been unaffected by the 1830–1780-Ma Capricorn Orogeny that deformed basins further to the west and records convergence of the Yilgarn and Pilbara cratons (Tyler and Thorne, 1990; Tyler et al., 1998; Pirajno and Adamides, 2000). Tyler et al. (1998) correlated deformation in the Stanley Fold Belt with the D2 phase of the Yapungku Orogeny that caused major folding and thrusting of the Rudall Complex to the north of the Earaheedy Basin at 1790–1760 Ma. These latter orogenic episodes may be part of a regional collisional event that resulted in the accretion and deformation of several Australian cratonic elements at ∼1780 Ma (Smithies and Bagas, 1997).

3. Sampling of the Frere Formation

3.1. Diamond drilling

In May 1999 we obtained oriented cores of unweathered iron-formation from the Frere Formation by drilling two holes, each 116 m deep, near the Gunbarrel Highway on Wongawol Station ∼200 km east of Wiluna (Fig. 1). Drilling was essential to obtain fresh samples of the Frere Formation from below the influence of Tertiary lateritisation and intense weathering that affects the rocks in the Earaheedy Basin to depths of 65 m or more.

The drilling took place in the southern part of the Earaheedy Basin, well outside the zone of deformation and metamorphism of the Stanley Fold Belt. The holes were drilled 6 km apart on opposite limbs of a northwest–southeast-trending gentle anticline to enable a fold test to be executed to constrain the timing of remanent magnetisation. Diamond drill hole (DDH) #1 (total depth 116.36 m) was sited on the northeastern limb of the anticline (26.33°S, 121.90°E, elevation 500 m) where outcropping beds dip at 10° toward 30°. DDH #2 (total depth 116.0 m) was sited on the southwestern limb of the anticline (26.34°S, 121.85°E, elevation 525 m) where outcrop dips at 20° toward 205°. DDH #1 was drilled with a hole dip angle of 70° toward 30° and DDH #2 with a hole dip angle of 70° toward 205°. (Note that true vertical depths are 94% of drilling depths given here because of the 70° dip of the drill holes.) HQ3 coring (60 mm core diameter) was carried out to 3–5 m depth, followed by NQ2 coring (50 mm core diameter) to total depth in DDH #1 and to 89.1 m in DDH #2; the latter hole was completed by NQ3 coring (45 mm core diameter). Core recovery was virtually 100% for DDH #1. Core recovery for DDH #2 was 50% in the weathered zone to 65 m depth and 88% for the remainder of the hole; core loss below the weathered zone was caused by broken ground.

The lowest point on the circumference of the core was determined at regular intervals during drilling by sliding a marking tool down each hole; the results implied that the strike of beds remained constant from the surface to total depth. The apparent dip of bedding as seen in the cores decreases by 10° on average over the full depth of each hole, although in places the decrease is as great as 5° over intervals of 1–3 m. Such changes in apparent dip of bedding are too abrupt to reflect changes in dip of the drill holes. The drilling characteristics while penetrating the dominant weathered shale/shale lithology indicated that these shallow holes did not deviate significantly (driller I. McKell, personal communication, 1999). Furthermore, as we observe in Section 5.2, the palaeomagnetic data for the drill cores are consistent with essentially straight holes. Hence, the overall decrease in apparent dip of bedding with depth is ascribed to flattening of the true dip of bedding, in part related to faulting and/or flexing of strata, with increasing depth and distance from the anticlinal axis.

The HQ and NQ cores were drilled out longitudinally in the laboratory to provide 25-mm-diameter core samples of all beds and lithologies of potential palaeomagnetic interest. Standard-sized core specimens (25 mm diameter × 20 mm high) for rock magnetic and palaeomagnetic analysis were prepared from these samples.

3.2. Lithology and mineralogy

The Frere Formation intersected by the drilling (Fig. 2) is reddened and lateritised near the surface and weathered to depths of 40 m in DDH #1 and 65 m in DDH #2. The unweathered rock comprises mainly

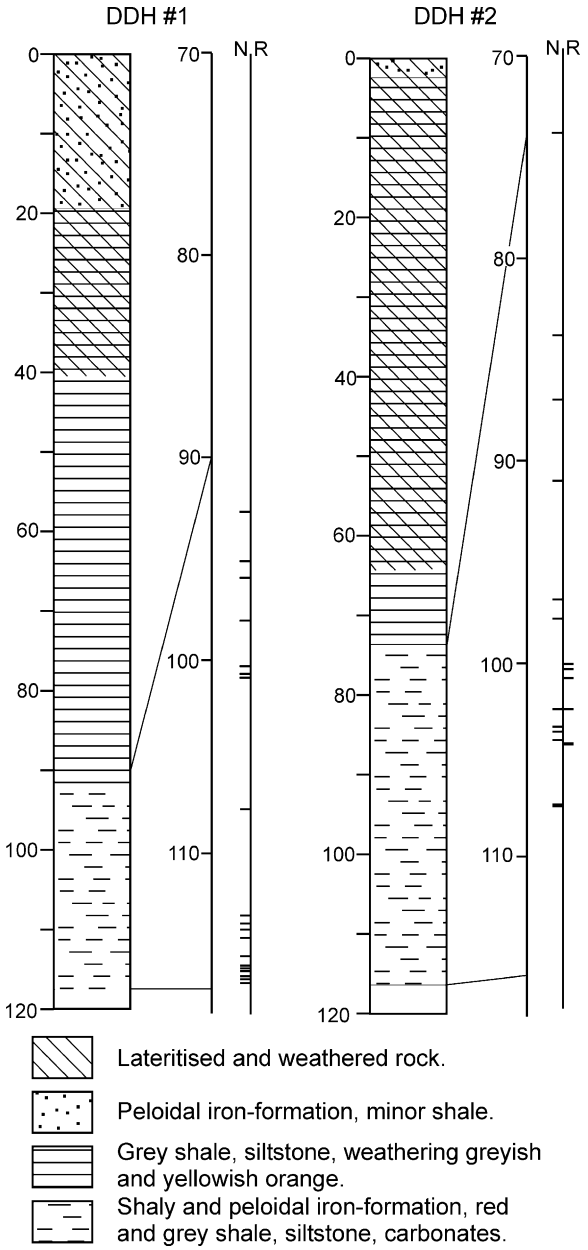


Fig. 2. Stratigraphy of the Frere Formation for DDH #1 and DDH #2, showing generalised lithologies, magnetostratigraphy, and depth of weathering. N, normal polarity; R, reverse polarity. Drilling depths are shown in metres for each hole drilled with a hole dip angle of 70°.

dark to medium grey siltstone and shale, as well as a distinctive facies comprising interbedded red and grey shale and siltstone, and peloidal ferruginous and cherty rocks including carbonates. The peloidal beds typically are graded and contain intraclastic material, suggesting that they are gravity deposits derived from a shallower part of the basin. The general similarity of the two sections suggests they may be partly correlative, with the section for DDH #2 extending ~17 m lower.

Iron-rich rocks from DDH #1 and DDH #2 and outcropping unmetamorphosed iron-formation in the southern part of the Earaheedy Basin (Hall and Goode, 1978; Goode et al., 1983; Bunting, 1986) contain hematite as ultrafine (<1 μm) dust, interpreted as primary. Peloidal rocks contain hematite as dense aggregates of small (<3 μm) platelets. With increasing metamorphism in the northern part of the basin, the ultrafine hematite is replaced progressively by platy and specular hematite, and magnetite. Other iron-bearing minerals identified in minor to trace amounts by X-ray diffraction analysis of samples from DDH #1 and DDH #2 include siderite, ankerite, stilpnomelane, and greenalite (latter tentative). This is the first identification of iron carbonates and iron silicates in the Frere Formation; their apparent absence in outcrop of the Frere Formation is ascribed to severe surface oxidation and prolonged weathering (Bunting, 1986).

Total Fe content of the different lithologies from DDH #1 and DDH #2, expressed as Fe_2O_3, is as follows: red shale and red siltstone, mean = 17.68%, range = 6.73–40.90% (16 samples); peloidal ferruginous rocks, including carbonates, mean = 17.60%, range = 2.23–32.91% (7); dark grey shale and siltstone, mean = 5.97%, range = 5.08–7.35% (9; data for the dark grey rocks from F. Pirajno, Geological Survey of Western Australia). The iron-rich rocks containing >15% Fe of primary origin (depositional or diagenetic) are iron-formation as defined by James (1966).

4. Rock magnetism

4.1. Laboratory techniques

Bulk susceptibility (k) was measured with a transformer bridge calibrated using paramagnetic salts

(Ridley and Brown, 1980). The bridge has a sensitivity of 10^{-6} SI. Anisotropy of magnetic susceptibility (AMS) was measured using a DIGICO anisotropy delineator interfaced to a computer. The operating system automatically corrects results for the DIGICO calibration error noted by Veitch et al. (1983) and Hrouda et al. (1983).

Thermomagnetic curves and susceptibility versus temperature measurements were obtained using a variable field translation balance designed and built at the University of Munich. This translation balance operates as a harmonic oscillator with damping in the forced oscillation mode. Fast Fourier transform of the signal enhances its sensitivity by orders of magnitude over conventional magnetic balances. The variable field translation balance can easily detect diamagnetic materials. A furnace/cooling unit allows samples to be measured between −175 and 700 °C. Samples are usually small cylindrical plugs (5 mm diameter × 5 mm high) although crushed samples (∼200 mg) also can be used if packed tightly enough. The small plugs were cored from off-cuts of standard-sized palaeomagnetic specimens.

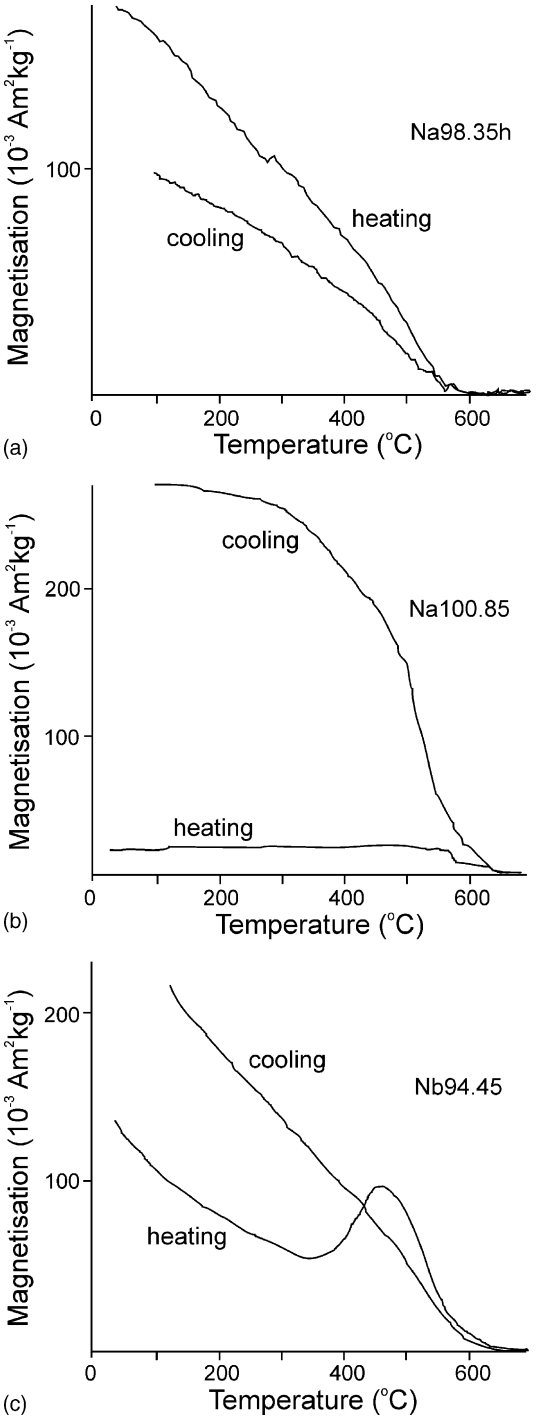
4.2. Results

AMS measurements for 86 specimens of shaly and peloidal iron-formation, red shale and red siltstone from DDH #1 and DDH #2 indicate that the Frere Formation is weakly anisotropic but exhibits a well-defined low *k* magnetic fabric dominated by a bedding-parallel foliation. There is no obvious indication of a tectonic fabric in the AMS data. Bulk susceptibilities are low (average $\sim 1 \times 10^{-3}$ SI, range 0.4×10^{-3} to 2.0×10^{-3} SI), indicating very low magnetite contents (average <0.05% by volume). The mean anisotropy ratios (major/minor axial susceptibilities) of 1.03–1.08 for these rocks, while low, may indicate the growth of some secondary (late diagenetic) magnetite resulting from subsurface fluid flow and burial pressures (≥4 km burial + erosion: see Table 1).

For comparison, AMS measurements were made for 44 specimens of peloidal iron-formation obtained from unoriented cores of flat-lying Frere Formation from Renison Gold Corporation vertical drill holes TDH-26 (25.55°S, 120.67°E; depths sampled = 149.5–383.5 m) and TDH-28 (25.55°S, 120.65°E; depths sampled = 187.0–188.3 m), drilled in the northwestern part of the Earaheedy Basin 25 km south of the Stanley Fold Belt (Fig. 1; samples provided by the Department of Mineral and Petroleum Resources, Perth). The results show that these rocks also have a bedding-parallel magnetic foliation but have higher susceptibilities (average 0.21 SI for TDH-26 and 0.74 SI for TDH-28) that indicate enhanced magnetite content (∼6% for TDH-26 and ∼20% for TDH-28). The high magnetite content may reflect alteration of the Frere Formation by the circulation of hydrothermal fluids related to deformation in the Stanley Fold Belt to the north. Anisotropy ratios also are substantially higher for specimens from these drill cores (1.21 for TDH-26 and 1.16 for TDH-28), suggesting growth of secondary magnetite in a stress field with vertical compression.

Susceptibility versus temperature measurements show that the predominant magnetic mineral in the Frere Formation from DDH #1 and DDH #2 is hematite but a small amount of magnetite also is present in some specimens. However, numerous specimens displayed clear evidence for the production of magnetite between 300 and 500 °C by alteration of a weakly magnetic phase. Such magnetite production mimics the well known response of siderite to heating in this temperature range (Pan et al., 2000), as evidenced by thermomagnetic analysis (see Section 5.2).

In broad terms, the behaviour of specimens falls into three groups. In the first two groups, specimens underwent little alteration until above the Curie temperature of magnetite (580 °C). Fig. 3a shows the thermomagnetic curve for specimen Na98.35h, from group 1, that altered slightly above the Curie temperature. Fig. 3b shows results for specimen Na100.85, from group 2, that altered quite substantially but only above the Curie temperature. Good palaeomagnetic results were obtained from these two groups. In the third group, significant alteration began well below the Curie temperature and spurious magnetisations acquired by the magnetite swamped any weak underlying hematite component. No sensible palaeomagnetic direction could be obtained from specimens of the third group. Specimen Nb94.45 is representative of group 3 and reveals magnetite production from below 400 °C (Fig. 3c).



5. Palaeomagnetism

5.1. Laboratory techniques

Routine palaeomagnetic laboratory methods (Collinson, 1983; Butler, 1992) were employed. Remanent magnetisations were measured using a 2G 755R three-axis cryogenic magnetometer interfaced to a computer. The demagnetisers used were a 2G 600 series alternating field (AF) demagnetiser and the CSIRO automated three-stage carousel furnace. The furnace is housed in a 4 m 10-coil Helmholtz set with automatic feedback maintaining zero-field (<5 nT). A batch of 18 specimens was run at the University of Western Australia's palaeomagnetic laboratory using a mu-metal shielded furnace. All samples were subjected to stepwise thermal demagnetisation. In addition, some samples were subjected to stepwise AF demagnetisation and AF pre-treatment prior to measuring after high-temperature thermal demagnetisation steps.

Components of magnetisation were isolated using an interactive version of Linefind (Kent et al., 1983), whereby linear segments are fitted to data points weighted according to the inverse of their measured variances. Because some specimens carry remanence of both normal and reverse polarities, indicating that remanence acquisition has spanned at least one geomagnetic reversal, acquisition has occurred over sufficient time to adequately average secular variation. Accordingly, we have assigned unit weight to specimens in the statistical analysis. Excluding premature directional instability evidently caused by the oxidation of siderite in many specimens, the highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature ob-

Fig. 3. Variation of magnetisation with temperature using a variable field translation balance and a field of 800 mT (8000 Oe) for: (a) specimen Na98.35h, showing minor alteration above the Curie temperature of magnetite (580 °C); (b) specimen Na100.85, showing considerable alteration confined to temperatures above the Curie temperature; and (c) specimen Nb94.45, showing alteration well below the Curie temperature. It was generally not possible to define a palaeomagnetic component from specimens that behaved like Nb94.45. Specimen prefixes Na and Nb refer to DDH #1 and DDH #2, respectively, and specimen numbers to drilling depths in metres.

served, is generally taken as an indication of the Curie temperature of the mineral carrying that component.

5.2. Results

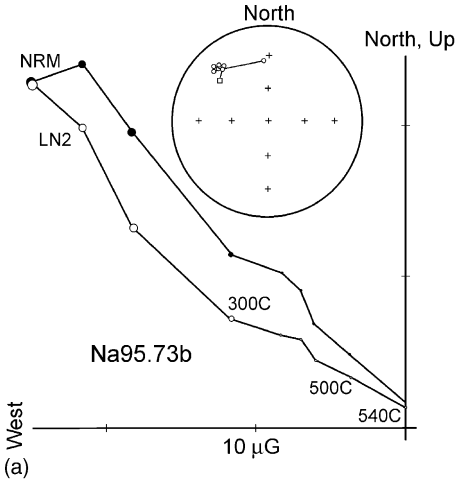
Natural remanent magnetisation (NRM) directions of specimens from the Frere Formation are clustered near the present Earth's magnetic field direction (declination $D = 1.5^\circ$, inclination $I = -60.6^\circ$), with an elongate tail extending across the northwest upper octant. The NRM mean direction is $D = 340^\circ$, $I = -61^\circ$, after normalising the polarities. Some samples from shallow depths displayed directions roughly anti-parallel to present field and probably record weathering when the field was of reverse polarity. NRM intensities depended on lithology: silty lithologies had a mean intensity of $10 \pm 6\,\mathrm{mA\,m^{-1}}$; for more shaly varieties the mean intensity was somewhat stronger at $21 \pm 12\,\mathrm{mA\,m^{-1}}$, probably reflecting the higher hematite content; and for peloidal rocks the intensities were variable and ranged from a few $\mathrm{mA\,m^{-1}}$ to $\sim 100\,\mathrm{mA\,m^{-1}}$, with a mean of $30 \pm 30\,\mathrm{mA\,m^{-1}}$. The distribution of NRM directions also can be interpreted in terms of an essentially northwest-directed component deflected toward the up-drill-hole directions. This would explain why the distribution bifurcates with one branch extending northeast steep-up and the other west-southwest steep-up. The drill holes had dip angles of 70° with respect to the horizontal and were aligned toward azimuths of 30 and 205°, which yields up-drill-hole directions of $D = 210^\circ$, $I = -70^\circ$, and $D = 25^\circ$, $I = -70^\circ$, respectively. Drilling-induced remanence preferentially affects multi-domain magnetite and is most easily eradicated using low-temperature demagnetisation (Schmidt, 1993).

After low-temperature demagnetisation, stepwise thermal demagnetisation was found to be the most effective method for resolving the magnetic components present and was carried out for 263 specimens. Many specimens acquired spurious components of magnetisation with cooling after heating to 300–500 °C. Such unstable magnetisation is ascribable to the presence of siderite, which oxidises to magnetite on heating (Pan et al., 2000). X-ray diffraction analysis confirmed the presence of siderite in samples of the Frere Formation that displayed this instability. The specimens that were thermally demagnetised using the University of Western Australia's shielded furnace also acquired spurious remanence on alteration, showing that such components are very difficult to avoid. We also experimented using an automatic AF demagnetisation step of 100 mT prior to measuring the remanence after high-temperature thermal demagnetisation steps, but found that the spurious components were extremely hard. Below the Curie temperature of magnetite these components are best ascribed to thermochemical remanent magnetisation acquired by single-domain magnetite as it forms from the breakdown of siderite, whereas above the Curie temperature the components are pure thermoremanent magnetisation. However, other specimens displayed stable, moderate- to high-temperature components, and inferred Curie temperatures of ~580 and 680 °C support the analysis by variable field translation balance that magnetite and hematite, respectively, are the principal carriers of remanence.

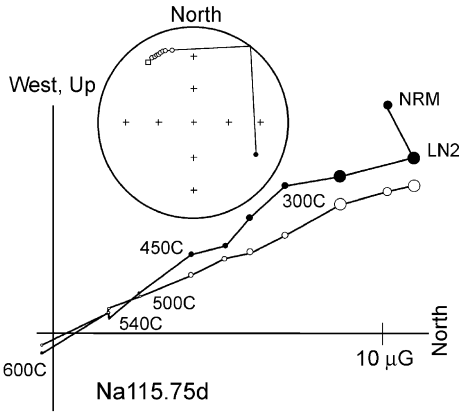
Palaeomagnetic data for the three components A, B and C recognised are summarised in Table 2 and data for components A and B for each drill hole are given in Table 3. Fig. 4 shows representative stepwise thermal demagnetisation orthogonal projections for components A and B. Directions of all components isolated using Linefind are plotted in Fig. 5.

Component C is carried by hematite. It shows both polarities (Fig. 5a) and is restricted to lateritised and weathered iron-formation that persists to 40 m depth in DDH #1 and 65 m depth in DDH #2. The in situ mean direction of $D = 183.5^\circ$, $I = 55.3^\circ$ ($K = 33.9$, $\alpha_{95} = 5.3^\circ$, $n = 23$) gives a palaeopole located at

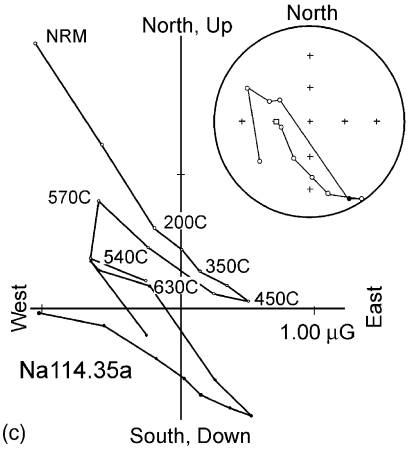
Fig. 4. Orthogonal projections showing examples of behaviour on stepwise thermal demagnetisation of specimens of unweathered Frere Formation, grouped according to the principal carriers of remanence: (a, b) magnetite; (c, d) magnetite and hematite; (e, f) hematite and trace magnetite. Closed circles represent magnetisation vectors projected onto the horizontal plane and open circles represent projections onto the vertical plane. Circle size indicates the circular standard deviation of each measurement. LN2 marks low-temperature demagnetisation, and some thermal demagnetisation steps also are labelled. Units are $\mathrm{mA\,m^{-1}}$ ($=1\,\mu\mathrm{G} = 10^{-6}\,\mathrm{emu\,cc^{-1}}$). Specimen prefix Na refers to DDH #1 and specimen numbers to drilling depths in metres.


North
North, Up
NRM
LN2
300C
Na95.73b
500C
540C
West
10 μG

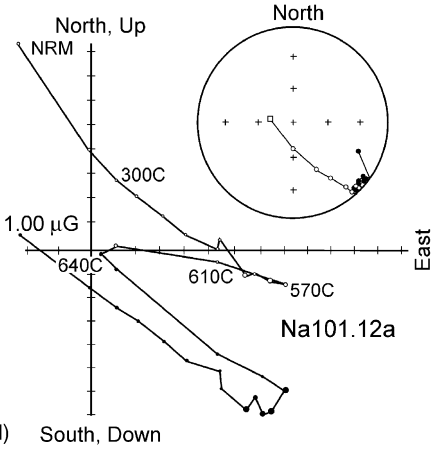
(a)


North
West, Up
NRM
LN2
300C
450C
500C
540C
North
10 μG
600C
Na115.75d

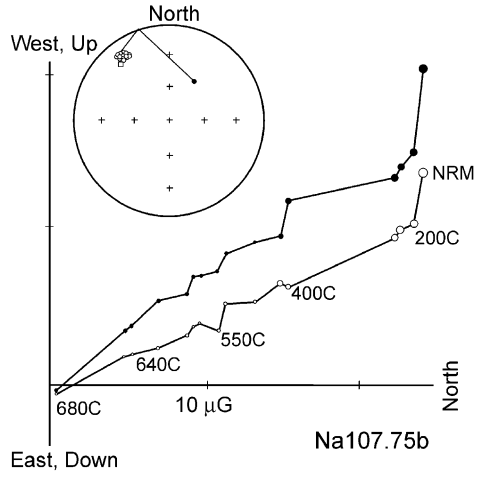
(b)


NRM
North, Up
North
570C
200C
540C
350C
630C
450C
West
East
1.00 μG
Na114.35a
South, Down

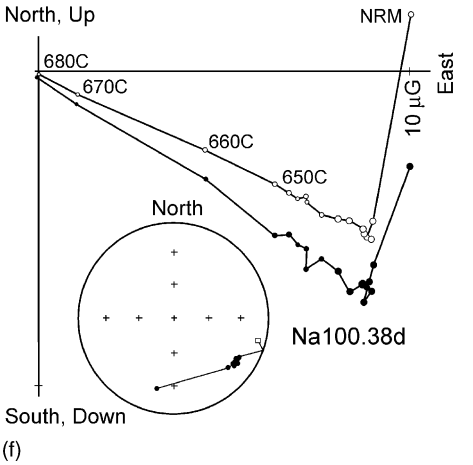
(c)


North, Up
North
NRM
300C
1.00 μG
East
640C
610C
570C
Na101.12a
South, Down

(d)


North
West, Up
NRM
200C
400C
550C
640C
680C
10 μG
North
Na107.75b
East, Down

(e)


North, Up
NRM
680C
670C
660C
650C
East
10 μG
North
Na100.38d
South, Down

(f)

Table 2
Summary of palaeomagnetic results for the Frere Formation, Earaheedy Basin, Western Australia

Component	Maximum T_{ub} (°C)	n	D_h (°)	I_h (°)	K	α_{95} (°)	D_b (°)	I_b (°)	K	α_{95} (°)
C	~680	23	183.5	55.3	33.9	5.3				
Pole C (in situ): latitude = 80.0°S, longitude = 105.3°E (dp = 5.3°, dm = 7.5°)										
B	~580	25	320.4	−26.6	46.2	4.3	320.4	−26.6	46.2	4.3
Pole B (in situ = bedding dip-corrected): latitude = 51.0°S, longitude = 222.2°E (dp = 2.5°, dm = 4.6°)										
A	~680	78	313.0	−25.6	37.1	2.7	314.3	−24.8	38.5	2.6[a]
Pole A (bedding dip-corrected): latitude = 45.2°S, longitude = 220.0°E (dp = 1.3°, dm = 2.4°)										
A + B		103					315.7	−25.2	39.0	2.3
Pole A + B (bedding dip-corrected): latitude = 46.5°S, longitude = 220.5°E (dp = 1.3°, dm = 2.4°)										

T_{ub}, unblocking temperature; n, number of specimens; D_h, I_h, mean declination and mean inclination with respect to present horizontal; D_b, I_b, mean declination and mean inclination with respect to bedding; K, precision parameter; α_{95}, semi-angle of cone of 95% confidence about mean direction; dp, dm, semi-minor and semi-major axes of polar error ellipse.
[a] Positive fold test at >99% confidence.

80.0°S, 105.3°E (dp = 5.3°, dm = 7.5°). Our results agree with palaeomagnetic data for lateritic profiles in the Perth Basin (Schmidt and Embleton, 1976) and are consistent with a mid-Miocene age for the lateritisation (see Idnurm, 1986). The Miocene remagnetisation of the weathered Frere Formation will be discussed in detail elsewhere.

Component B (Figs. 4a, b and 5b) is carried by magnetite and has a mean direction of $D = 320.4°$, $I = -26.6°$ ($K = 46.2$, $\alpha_{95} = 4.3°$, $n = 25$) that gives a palaeopole located at 51.0°S, 222.2°E (dp = 2.5°, dm = 4.6°). There is no bedding dip correction for the B component, which occurred in samples from a depth where beds are flat-lying. Component B appears to be entirely of normal polarity (northwest-up), although some reverse directions (southwest-down) also share unblocking temperatures below the Curie temperature of magnetite. It is therefore possible that some of the reverse component is carried by magnetite. However, in most cases the reverse component is associated with unblocking temperatures clearly higher than that of magnetite and generally up to 670–680 °C. This finding suggests that the reverse component is carried predominantly by hematite.

Component A (Figs. 4c–f, 5c and d) was identified for shaly and peloidal iron-formation, red shale and red siltstone, and shows both polarities. Before combining the directions from DDH #1 and DDH #2, their means were tested for a common mean direction (McFadden and Lowes, 1981). The in situ comparison shows that the means do not share a common mean and are statistically different. The McFadden–Lowes test statistic $f = 3.731$, compared to the 0.05 cut-off $F(2, 36) = 3.258$. After correcting for bedding dip, however, the means do share a common mean and are not statistically different. The McFadden–Lowes test statistic $f = 1.209$, compared to the 0.05 cut-off $F(2, 36) = 3.258$. This finding justifies combining the bedding dip-corrected A directions from DDH #1 and DDH #2, and constitutes a positive fold test (McFadden and Jones, 1981).

The A component also passes the McFadden (1990) fold test with >99% confidence. In situ $\xi_1 = 22.51$ compared to critical values at 95% confidence of 10.27 and 99% confidence of 14.53 (where ξ_1 is the test statistic (definition 1) for correlation of bedding dip azimuths with the displacement direction of unit vectors from their mean direction; McFadden, 1990). The bedding attitude and magnetisation directions appear to be correlated, so the magnetisation seems not have been acquired in the present orientation. After correcting for dip of bedding, the directions are distributed so they no longer correlate with the dip azimuths, in other words their displacement from the mean direction is random, with $\xi_1 = 9.60$. This result is consistent with the magnetisation being acquired in this orientation, i.e. pre-folding.

A reversal test (McFadden and McElhinny, 1990) performed on component A is negative at $P = 0.05$, both for in situ directions and for bedding dip-corrected directions. The in situ directions fail at $P = 0.01$ (i.e. with 99% confidence), whereas the

Table 3
Palaeomagnetic data, grouped by depth, for Frere Formation components A and B for each drill hole (DDH #1 and DDH #2), Earaheedy Basin

Hole and depth (m)	Bedding dip (°)	Component	*n*/*N*	D_h (°)	I_h (°)	D_b (°)	I_b (°)	*K*	α_{95} (°)
DDH #1									
92	0	A	3	318.1	−46.4	318.1	−46.4	22.5162	26.6
95	0	A	6	315.7	−18.9	315.7	−18.9	33.3610	11.6
96	0	A	2	308.1	−36.1	308.1	−36.1	43.0693	39.0
98	0	A	4	321.6	−30.6	321.6	−30.6	140.541	7.8
100	0	A	4	313.5	−24.1	313.5	−24.1	31.1047	16.7
101	0	A	3	310.3	−34.5	310.3	−34.5	51.3828	17.4
107	0	A	2	323.8	−19.8	323.8	−19.8	1911.72	5.7
112	0	A	2	309.2	−27.3	309.2	−27.3	19.3925	60.2
113	0	A	9	315.8	−27.3	315.8	−27.3	63.4938	6.5
114	0	A	2	328.6	−28.1	328.6	−28.1	21.9407	56.2
115	0	A	7	319.4	−19.2	319.4	−19.2	117.524	5.6
116	0	A	7	314.3	−24.9	314.3	−24.9	167.884	4.7
Mean		A	12	316.6	−28.2	316.6	−28.2	70.7715	5.2
DDH #2									
73	15	A	2	317.3	−36.3	326.0	−29.5	55.6861	34.1
90	10	A	3	307.5	−26.4	312.0	−23.8	43.2431	19.0
97	12	A	2	302.2	−13.9	305.0	−12.1	51.4093	35.6
100	10	A	5	300.4	−20.7	304.0	−19.4	89.2120	8.1
102	8	A	2	312.0	−29.4	316.0	−26.8	57.0289	33.7
103	9	A	7	308.7	−16.5	311.1	−14.2	46.6663	8.9
104	10	A	3	296.9	−24.1	301.3	−23.4	132.092	10.8
107	5	A	3	315.6	−34.2	318.6	−26.1	320.054	6.9
Mean		A	8	307.3	−25.3	311.5	−22.9	61.0692	7.1
Mean for both holes		A	20	312.8	−27.1	314.5	−26.1	62.1272	4.2
(For mean D_h, I_h: $K = 58.5036$, $\alpha_{95} = 4.3$)									
DDH #1									
92	0	B	3	340.8	−40.2	340.8	−40.2	133.081	10.7
95	0	B	7	318.8	−27.4	318.8	−27.4	73.1305	7.1
98	0	B	3	323.3	−28.9	323.3	−28.9	272.225	7.5
113	0	B	4	312.3	−19.2	312.3	−19.2	62.8043	11.7
115	0	B	3	325.2	−22.3	325.2	−22.3	373.175	6.4
116	0	B	5	314.6	−23.4	314.6	−23.4	136.354	6.6
Mean		B	6	321.9	−27.2	321.9	−27.2	50.2929	9.5

DDH #1 azimuth = 30°, hole dip angle = 70°. DDH #2 azimuth = 205°, hole dip angle = 70°. Bedding dip is relative to the horizontal. Component A carried by hematite, maximum $T_{ub} \approx 680$ °C. Component B carried by magnetite, maximum $T_{ub} \approx 580$ °C. *n*/*N*: number of specimens grouped at a given depth/number of groupings of specimens. Other abbreviations as for Table 2.

bedding dip-corrected directions pass if *P* is relaxed to 0.04. This result may be interpreted as further evidence that component A was acquired prior to Stanley folding. The asymmetry is not easily explained as contamination by an unrecognised secondary component, because the best-fitting great circle joining the overall mean with the mean normal and mean reverse directions does not pass near either the present field direction or the up-drill-hole directions (however, these possible secondary directions do appear to be recorded in the NRM data). We therefore interpret the asymmetry to indicate a small, although significant, departure of the geomagnetic field from a geocentric axial dipole. Asymmetry of reversals is known from other Precambrian rocks such as the 1.1-Ga Keweenawan volcanics and intrusives (Pesonen and Nevanlinna, 1981) and the ~1.9-Ga Gunflint Formation in the Lake Superior region (Schmidt and

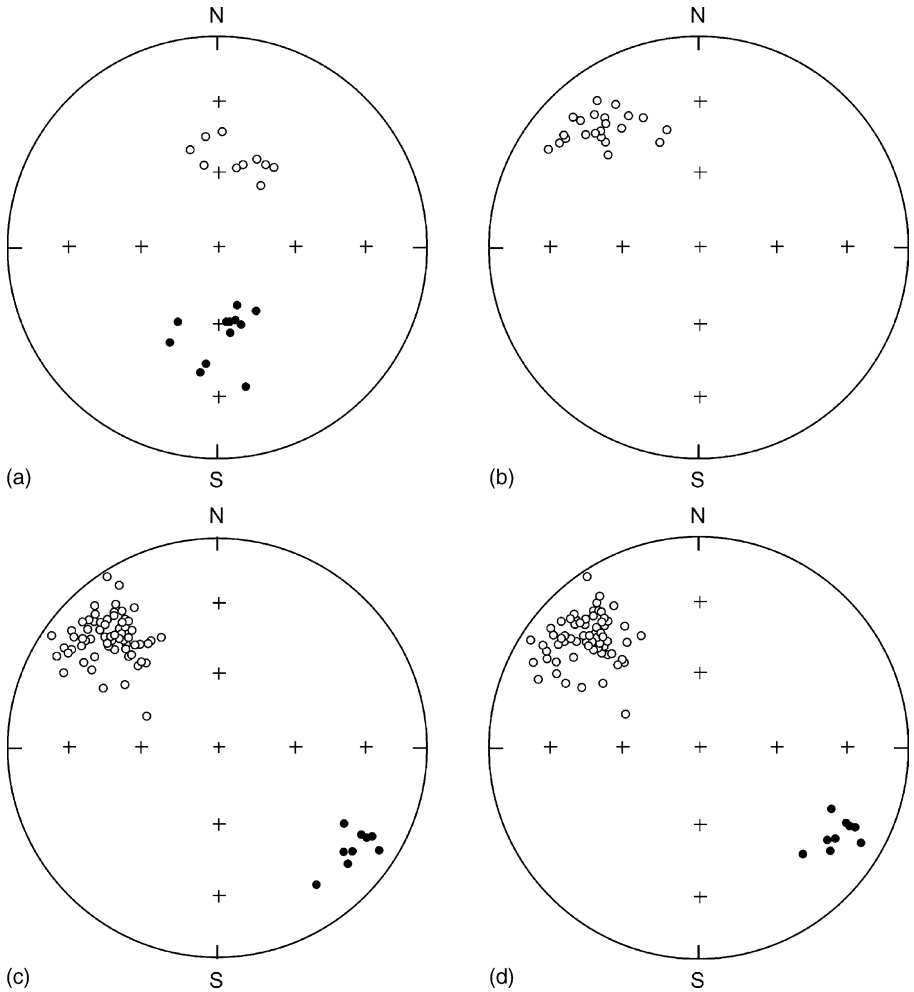


Fig. 5. Equal-area stereographic projections of directions for: (a) high unblocking temperature component C (in situ, maximum $T_{ub} \approx 680\,^{\circ}C$) for lateritised Frere Formation; (b) lower unblocking temperature component B (in situ, maximum $T_{ub} \approx 580\,^{\circ}C$) for unweathered Frere Formation; (c) high unblocking temperature component A (in situ, maximum $T_{ub} \approx 680\,^{\circ}C$) for unweathered Frere Formation; (d) component A (bedding dip-corrected). Closed symbols indicate down directions and open symbols up directions.

Williams, 2002), and has been interpreted in terms of persistent (zonal) non-dipole fields. This issue will be explored elsewhere.

Polarity reversals displayed by component A in the stratigraphic section for DDH #2 define a magnetostratigraphy that is not duplicated in the section for DDH #1 (Fig. 2), which is consistently of normal polarity and, as noted in Section 3.2, may be at a slightly higher stratigraphic level. Despite the asymmetry of the reversal, implying the presence of a minor non-dipole component, the presence of polarity reversals in one stratigraphic section and of mixed polarities within some specimens argues for the identification of a dipole field axis and the sufficient averaging of secular variation to define a palaeopole for the Frere Formation. The bedding dip-corrected A direction is $D = 314.3^{\circ}$, $I = -24.8^{\circ}$ ($K = 38.5$, $\alpha_{95} = 2.6^{\circ}$, $n = 78$), which gives a palaeopole located at 45.2°S, 220.0°E (dp = 1.3°, dm = 2.4°).

The red shale and iron-formation from the Frere Formation and their mineralogy accord with features

of 'Type A red beds' (Turner, 1980), for which the mineralogy at the time of deposition is fairly mature and the remanence is an early CRM. Hence the acquisition of remanence prior to Stanley folding, the presence of a magnetostratigraphy, and the nature of the hematitic sediments argue that component A is a CRM acquired soon after deposition. Component B is carried by trace amounts of magnetite of possible late diagenetic origin resulting from subsurface fluid flow and burial pressures. While A and B directions are statistically close, we prefer to use the hematite A result in palaeogeographic applications because it passes the fold test. Table 2 gives the combined result A + B for completeness.

The palaeomagnetic data imply that the drill holes show no significant deviation. Statistically constant remanence directions for components A and B would be expected over the depth intervals and area we sampled. Importantly, data for both drill holes (Table 3) show no discernible trends in declination and inclination over intervals of 24–34 m—if the drill holes had deviated substantially the remanence directions would not be so consistent.

6. Palaeogeographic and tectonic implications

The palaeomagnetic findings carry important implications for the palaeogeographic and tectonic settings of the Earaheedy Basin during and following iron-formation deposition.

6.1. Palaeolatitude of iron-formation deposition

Component A, here ascribed to a CRM acquired soon after deposition, carries information on the palaeolatitude of the Frere Formation. Employing the dipole equation, $\tan I = 2\tan\lambda$, where I is magnetic inclination and λ is geographic (palaeo)latitude (Butler, 1992), component A implies deposition of the Frere Formation at a palaeolatitude of $13.0 \pm 1.5^{\circ}$. To our knowledge, this is the best-constrained palaeolatitude directly determined for a Palaeoproterozoic iron-formation.

6.2. Tectonic setting at the Yilgarn–Pilbara boundary

The palaeomagnetic poles for components A and B for the Frere Formation are plotted on a global projection in Fig. 6 together with selected late Palaeoproterozoic poles from the Hamersley Province of the southern Pilbara and from the North Australian Craton. Pole A is corrected for dip of bedding, whereas pole B is from flat-lying beds. Poles A and B fall near many poles for 1.8–1.7 Ga from the West Australian and North Australian cratons, lending support to the hypothesis that these cratons have been joined at least since that time (Li, 2000).

In detail, poles A and B are closest to, but still significantly different from, poles for synfolding and postfolding overprints from the Hamersley Province that are ascribable to the Capricorn Orogeny, which peaked at ~1830–1780 Ma (Tyler and Thorne, 1990; Tyler et al., 1998) and was the last major orogeny to significantly affect the Hamersley Province (Li, 2000). These overprint poles are the synfolding pole from the Mount Jope Volcanics (Schmidt and Embleton, 1985), the in situ pole from iron ores at Paraburdoo (Schmidt and Clark, 1994), and the in situ pole from iron ores at Tom Price (Schmidt and Clark, 1994). However, vertical-axis relative anticlockwise rotation of 10–15° for Frere pole A brings it close to the Pilbara overprint poles, a finding that carries implications for the tectonic setting and deformation of the Earaheedy Basin.

The data imply that the Earaheedy Basin and possibly the Yilgarn Craton underwent clockwise rotation of 10–15° relative to the Pilbara Craton during Stanley folding near 1.78 Ga. This finding accords with the occurrence of tectonic settings marked by sinistral strike-slip faulting along the northern margin of the Yilgarn Craton during deposition of the Earaheedy Group (Myers et al., 1996) and evidence that closure of the Earaheedy Basin during Stanley folding involved sinistral strike-slip deformation adjacent to the Yilgarn–Pilbara cratonic boundary (Krapez and Martin, 1999).

6.3. Basin orientation

Information on basin or seaway orientation could shed light on the palaeogeographic setting and distribution of Proterozoic iron-formations. The original orientation of basins and ocean margins may have influenced oceanic circulation patterns such as the upwelling of deep waters and the movement of hydrothermal plumes, which are a probable source of

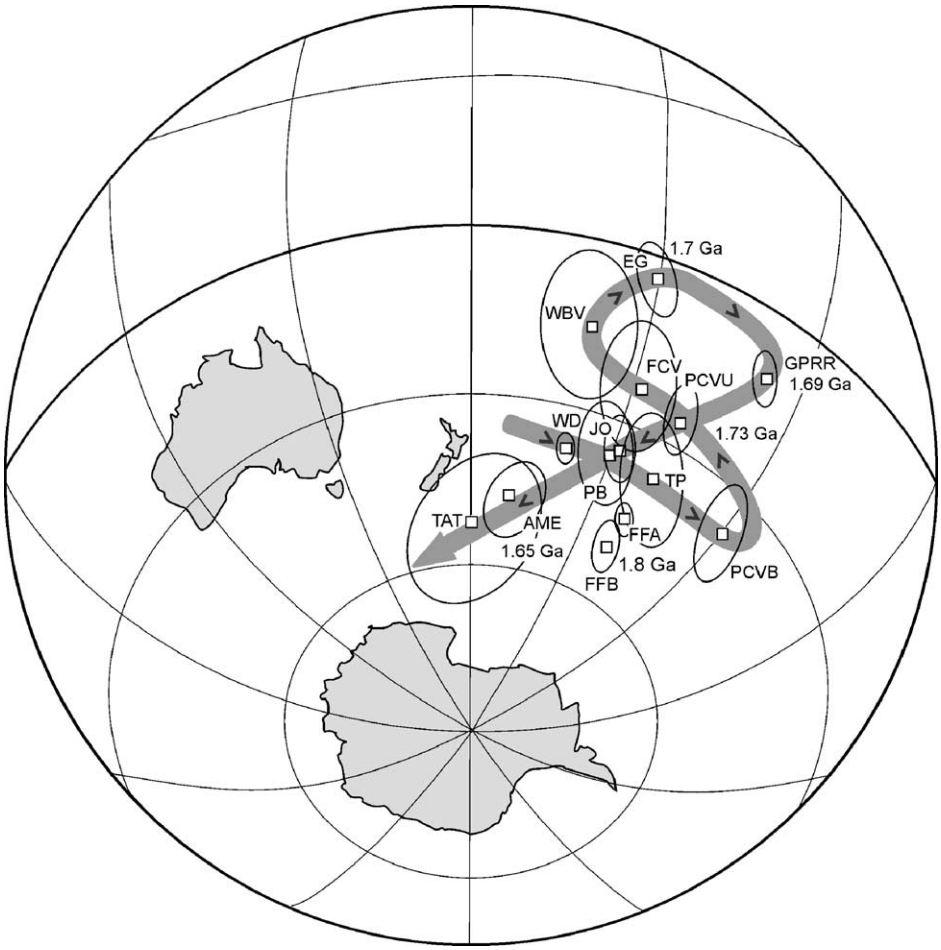


Fig. 6. Palaeomagnetic poles for Frere Formation components A and B (FFA and FFB, this study), late to post-Capricorn Orogeny overprint poles from the Hamersley Province of the southern Pilbara (WD: Wittenoom Dolomite, Li et al., 1993; JO: Mount Jope Volcanics, Schmidt and Embleton, 1985; PB: Paraburdoo hematitic ore body, Schmidt and Clark, 1994; TP: Tom Price hematitic ore body, Schmidt and Clark, 1994), and other late Palaeoproterozoic poles for Australia (EG, Elgee Formation, Li, 2000; PCVB, basalt near base of Peters Creek Volcanics; PCVU, upper Peters Creek Volcanics; FCV, Fiery Creek Volcanics; WBV, West Branch Volcanics; GPRR, Gunpowder Creek Formation; AME, Amelia Dolomite; TAT, Tatoola Sandstone, all from Idnurm, 2000). Polarities are not implied.

Palaeoproterozoic iron-formations and which today are driven preferentially to the west (Isley, 1995).

Component A for the Frere Formation implies a palaeo-northwest–southeast orientation for the Earaheedy Basin during iron-formation deposition (Fig. 7). The palaeo-orientation shown was determined after vertical-axis relative anticlockwise rotation of 12° for Frere palaeopole A to bring it close to penecontemporaneous Pilbara overprint poles (see Section 6.2). The geographic polarity of the A direction is unknown, however, because of the incompleteness of the Australian Proterozoic apparent polar wander path. Hence the present data cannot indicate whether the distal, deeper portion of the Earaheedy Basin from which upwelling waters or hydrothermal plumes may have been derived (see Pirajno and Adamides, 2000) then lay to the palaeo-northeast or palaeo-southwest of the nearshore basin margin adjacent to the Yilgarn Craton. If by further palaeomagnetic work Australia's position in Proterozoic supercontinents can be better constrained, however, we could transfer the apparent polar wander paths of other continents to Australia.

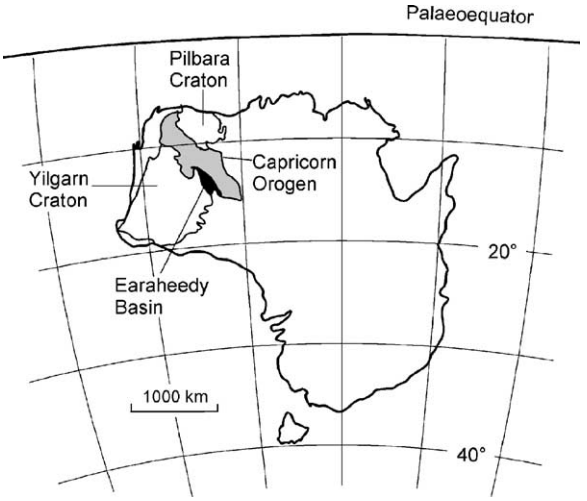


Fig. 7. Palaeo-orientation of the Earaheedy Basin and Capricorn Orogen during deposition of the Frere Formation, showing palaeolatitudes for Frere component A after vertical-axis relative anticlockwise rotation of 12° for Frere palaeopole A to bring it close to penecontemporaneous Pilbara overprint poles (geographic polarity unknown).

Through this bootstrap approach the palaeogeographic settings of sedimentary basins, including basin orientation, might become better understood.

7. Conclusions

The main findings of our study are as follows:

1. Unmetamorphosed shaly and peloidal iron-formation from the Frere Formation (1.9–1.8 Ga) in the Earaheedy Basin at the northeastern margin of the Yilgarn Craton yielded a stable, high-temperature (680 °C), dual-polarity palaeomagnetic component A with a bedding dip-corrected direction of $D = 314.3°$, $I = -24.8°$ ($K = 38.5$, $\alpha_{95} = 2.6°$, $n = 78$). A fold test was positive at >99% level of confidence, indicating acquisition of remanence prior to Stanley folding that occurred at ~1.78 Ga near the end of Capricorn orogenesis. Component A is ascribed to a CRM acquired soon after deposition.
2. Component A defines a palaeopole for the Frere Formation located at 45.2°S, 220.0°E (dp = 1.3°, dm = 2.4°), and implies a palaeolatitude of 13.0 ± 1.5° for the Frere Formation.
3. Frere pole A plots adjacent to, but is still significantly different from, late to post-Capricorn Orogeny overprint poles for the Hamersley Province of the southern Pilbara. However, vertical-axis relative anticlockwise rotation of 10–15° for Frere pole A brings it close to the Pilbara overprint poles.
4. The data imply that during Stanley folding the Earaheedy Basin and possibly the Yilgarn Craton underwent clockwise rotation of 10–15° relative to the Pilbara Craton. Such relative rotation accords with evidence for tectonic settings marked by sinistral strike-slip deformation adjacent to the Yilgarn–Pilbara cratonic boundary during deposition and folding of the Earaheedy Group. This deformation may mark the final stage of consolidation of the West Australian Craton and its convergence with the North Australian Craton.
5. The location of Frere pole A is consistent with the West Australian and North Australian cratons being in close proximity since at least ~1.8 Ga.
6. During deposition of the Frere Formation the Earaheedy Basin was oriented palaeo-northwest–southeast, although the geographic polarity is unknown.

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